

Robust Charging Infrastructure Design under Demand Uncertainty

A solar-powered charging station for a shared e-micromobility fleet

Which station do you build when the future is uncertain?

Theme 3 — solar charging station

Theme 2 — charge/discharge profiles

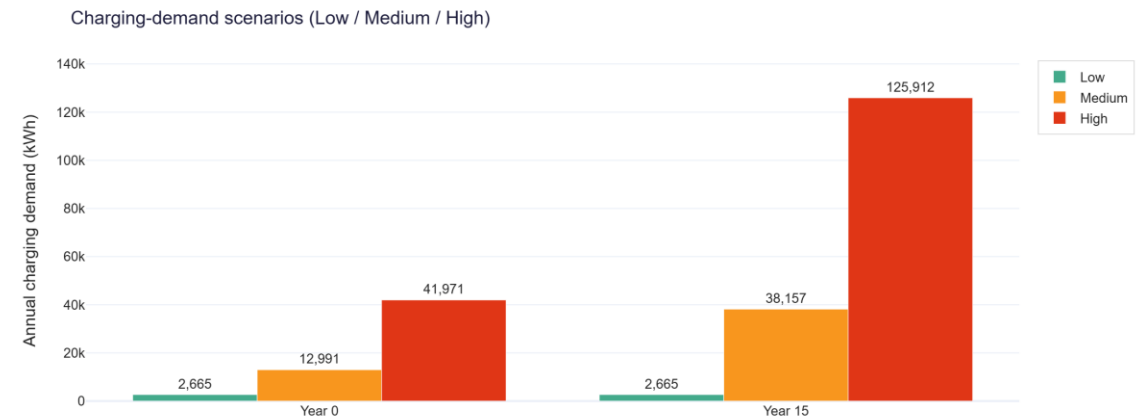
The problem: sized for average, or for the worst?

A mixed-fleet depot. The hub charges e-scooters, e-bikes and e-cargo bikes; demand swings with season, weather, events and growth.

The dilemma. Size for average demand and the fleet is stranded at peaks; size for the worst case and capital is wasted.

A constrained grid. A higher import limit needs a costly DNO reinforcement, so the evening peak must come from on-site solar + storage.

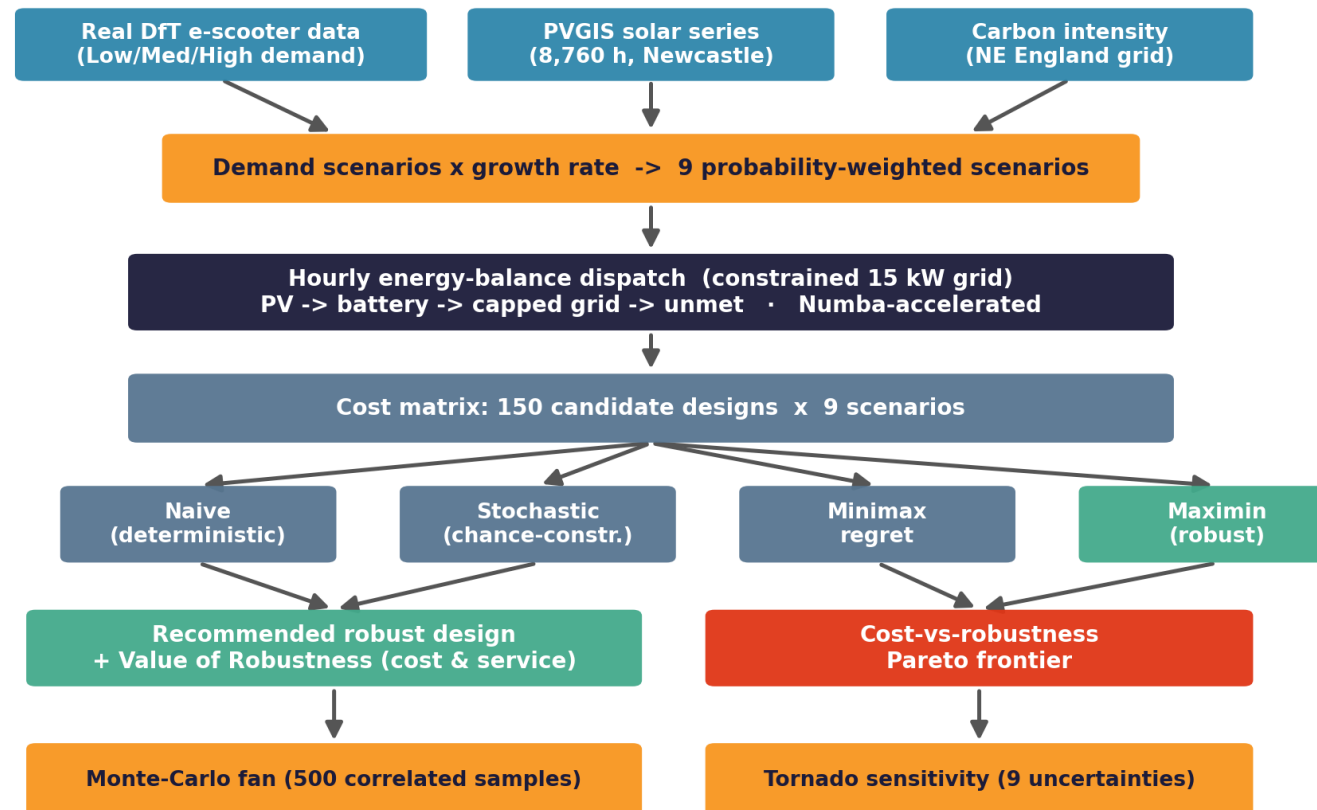
Real data. Low / Medium / High scenarios are built from the real DfT Newcastle e-scooter trial — trips, fleet, deployment, distance.



Demand scenarios derived from real trial data.

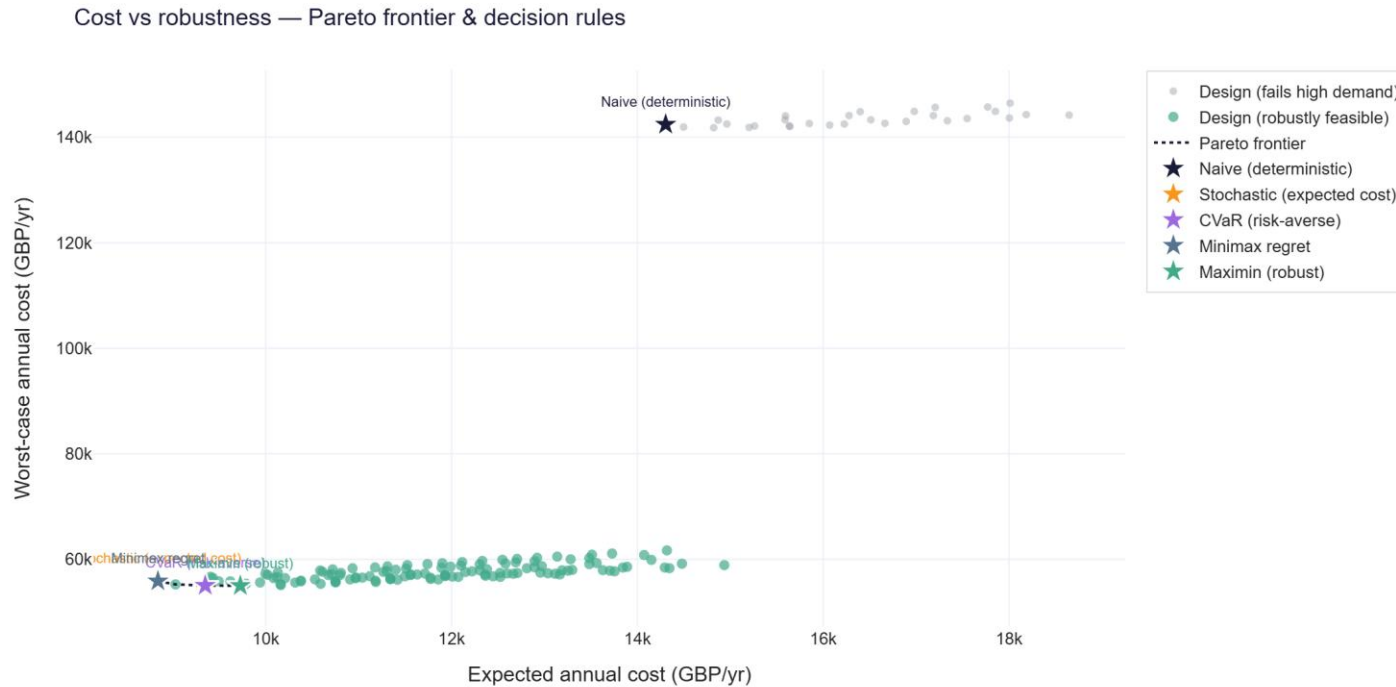
Approach — a reproducible robust-design pipeline

Methodology — Robust Solar Charging-Station Design



Every stage is reproducible from one command (run_analysis.py); all assumptions traceable in config.py

The result: robustness pays



-61%

worst-case annual cost vs the naive design

85% -> 96%

guaranteed fleet service, worst demand

£55,002/yr

robust worst-case cost (down from £142,423)

A small expected-cost premium buys large protection against the demand tail.

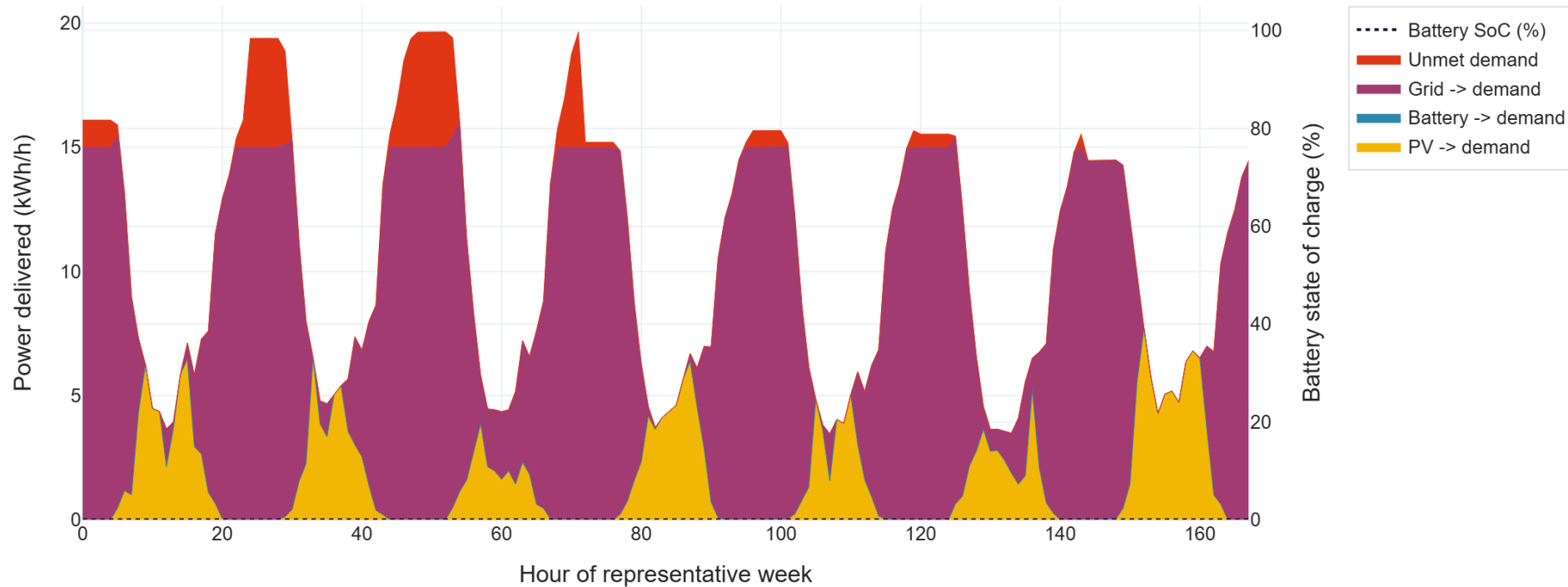
Smart charging & the recommended design (Theme 2)

20 kWp solar PV

8 charge bays, smart-managed

battery only if grid < ~10 kW

Hourly smart-charging dispatch — PV / battery / grid (Theme 2)



Smart control schedules flexible charging into sunny / off-peak hours and flattens the overnight load below the connection limit — no battery needed at a connected depot.

Feasible today — and sustainable

~£36,000

capital cost — off-the-shelf PV, Li-ion & AC chargers

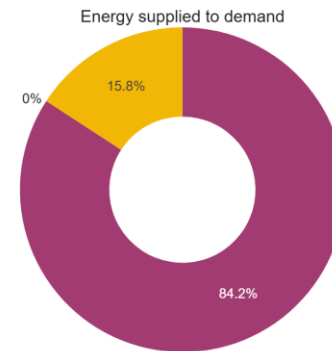
-16% CO₂

vs grid-only charging (4.6 tCO₂/yr saved)

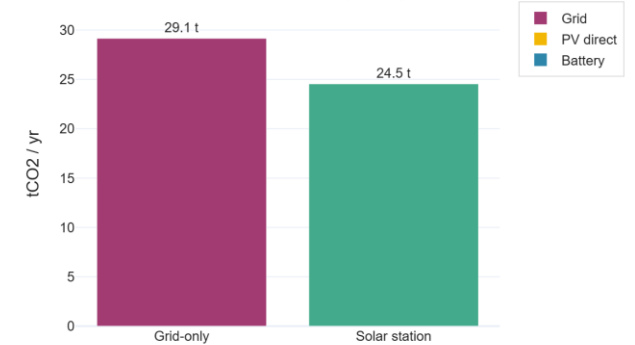
7/7 validated

key outputs within published benchmark ranges; 1-command rebuild

Energy mix & carbon savings (Theme 3 sustainability)



Annual CO₂: station vs grid-only



Original, authentic, reproducible

Every line of code and every figure is original, written from scratch. All three datasets are real: DfT e-scooter data (Open Government Licence), live PVGIS solar and National Grid ESO carbon intensity. Fixed seed, one-command rebuild, passing tests — built for the AI and plagiarism checks.



**We turn demand uncertainty from a risk
into a design input — and show that robustness pays.**

Thank you.